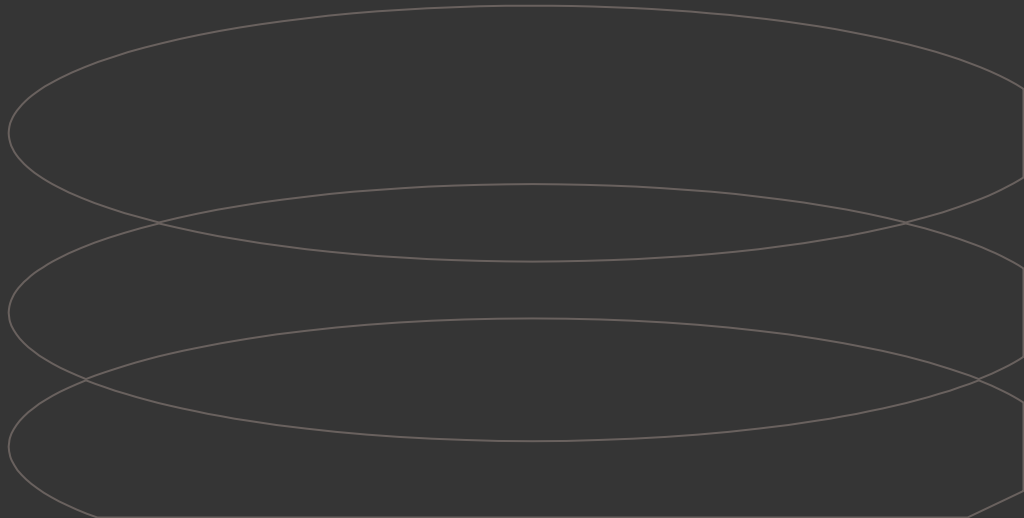
Georgios Koiliarakis - Qazi Danial

Rainbow ColorMaps and there stigmas

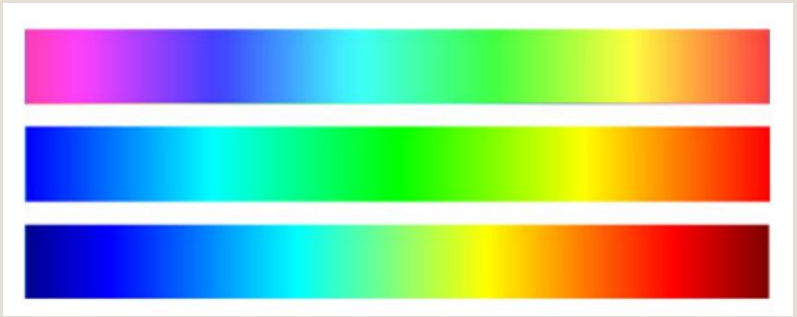
The usefulness of rainbow colormaps and gradients in information visualization

What is this article about?

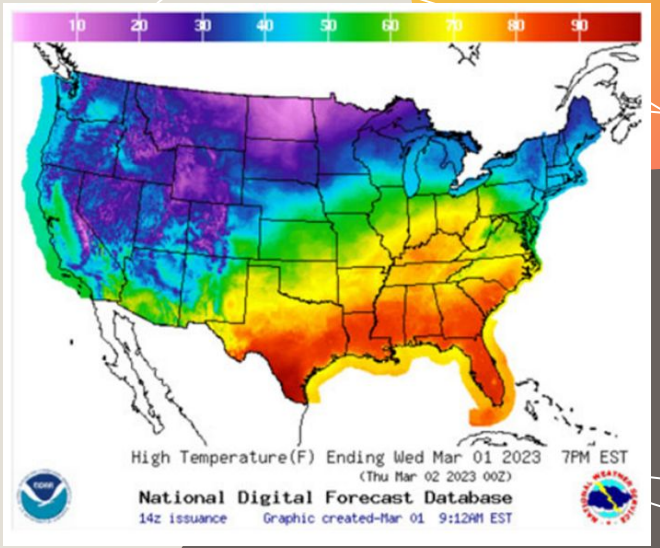


What are Rainbow Colormaps

A sequence of colors used to map data values onto an image.



Most of us have seen this type of rainbow colormap, which reflects how our minds perceive weather forecasts.



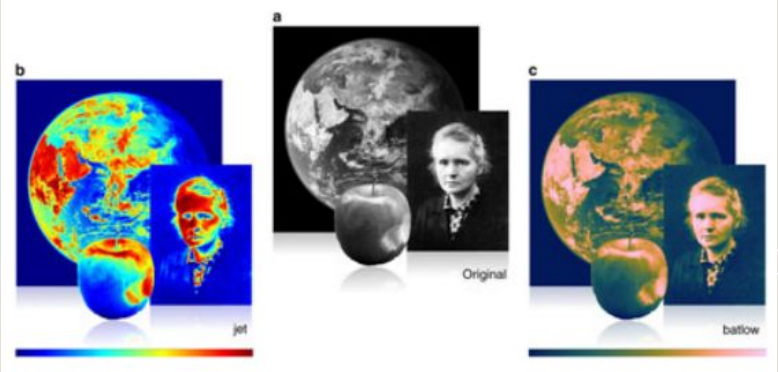
Criticism & Dogmatic Opposition

Nonmonotonic Luminance

- Shape of a surface can be inferred from luminance variation
- Shape-from-shading

Perceptual Nonuniformity

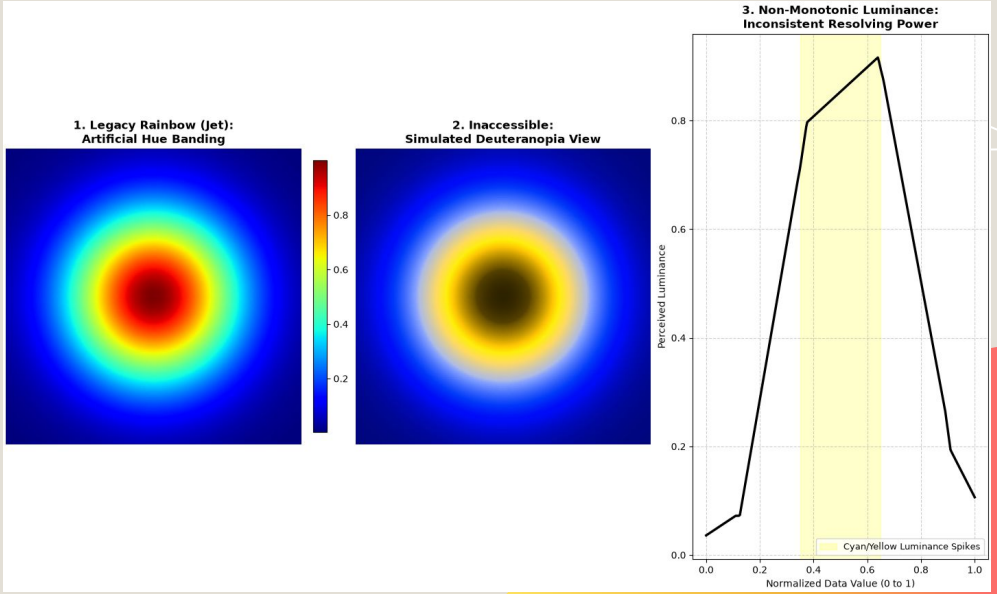
- JNDs (smallest difference b/w two colors)
- More JNDs in successive colors can help resolve more features in data
- Factors: colors, surrounding region
- CIE L* Profile



Accessibility

Problems arise with hue banding

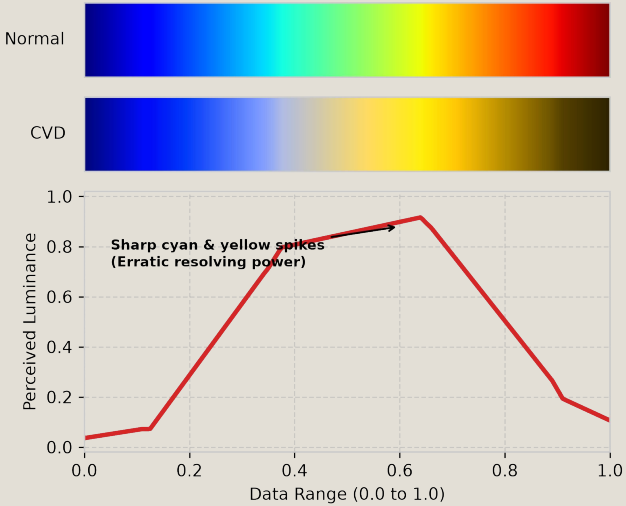
- sharp separate "steps"
- Color separation
- erratic across the data range.



The Solution:
Difference between colormaps

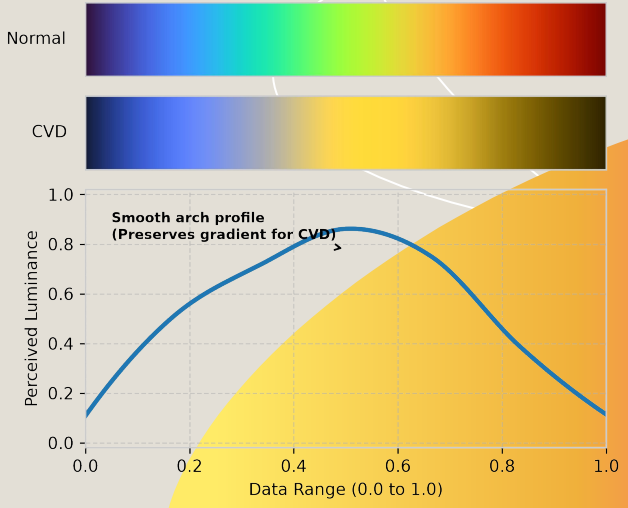
Legacy Rainbow (HSV / Jet)

More erratic, sharp drops and spikes in luminance that hide real features and invent false contours.



Turbo & Paraview Uniform

Feature smooth, intentionally designed luminance profiles. Better for accessibility with color ranges



Conclusion

- Not universally harmful
- Problems come from poor luminance & perceptual design
- Modern rainbow variants (Turbo, Batlow, Paraview) fix major issues
- Remain useful for specific tasks
- Avoid dogma, choose colormaps based on task, data and perceptual goals
- Effective if designed well and use in right context

